Siva Sai[1], Ishika Goyal[1], Vinay Chamola[1], and Rajkumar Buyya[1]

[1]Affiliation not available

November 05, 2025

Posted on 5 Nov 2025 — CC-BY 4.0 — https://doi.org/10.36227/techrxiv.176231624.49647500/v1 — e-Prints posted on TechRxiv are preliminary reports that are not peer reviewed. They should not b...

# Generative AI in the Age of Quantum Computing: A Taxonomy, Architectural Elements and Future Directions

Siva Sai*, Ishika Goyal, Vinay Chamola *Senior Member, IEEE*, Rajkumar Buyya *Fellow, IEEE*

***Abstract*—Generative AI has emerged as a transformative paradigm for diverse applications, yet the escalating scale of modern models exposes critical computational and memory bottlenecks in classical hardware. This paper investigates the intersection of quantum computing and generative artificial intelligence (QGAI) to address these limitations and scale modern generative models. As models grow to billions of parameters, classical systems face bottlenecks in memory, energy, and training efficiency, while quantum computing offers exponential representational benefits for high-dimensional data. The paper analyzes five core quantum generative architectures - Quantum Circuit Born Machines, Quantum Generative Adversarial Networks, Quantum Boltzmann Machines, Quantum Variational Autoencoders, and Quantum Diffusion models, highlighting their design principles, learning mechanisms, and applications. QGAI models have demonstrated significant promise in domains such as drug discovery, human–machine interaction, IoT security, and financial modelling. Despite these advances, QGAI remains constrained by qubit noise, barren plateaus, and integration challenges. We conclude by identifying ten open research challenges and propose directions for achieving scalable, interpretable, and energy-efficient quantum generative learning.**

***Index Terms*—Quantum computing, Generative AI, Quantum GANs, Quantum Boltzmann Machines, Quantum autoencoders, Diffusion models, NISQ, Quantum Machine Learning, Quantum advantage, Quantum applications.**

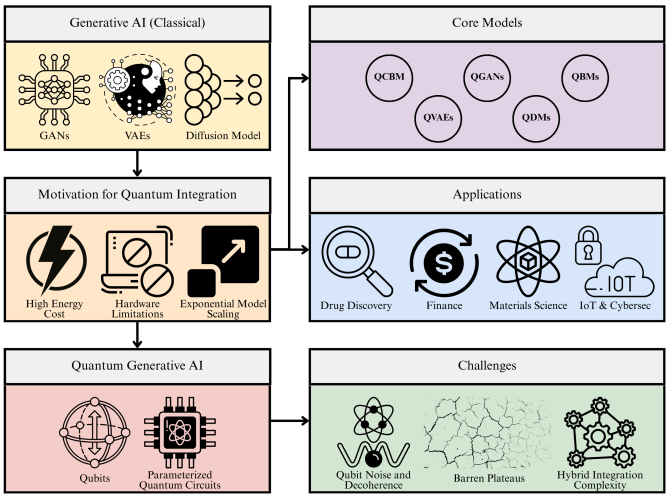


Figure 1: Landscape of Quantum Generative Artificial Intelligence (QGAI)

## I. Introduction

Generative artificial intelligence (GAI) has become a cornerstone of modern machine learning and creative automation. From Generative Adversarial Networks that synthesise photorealistic imagery through Variational Autoencoders that learn disentangled latent spaces [1] to the recent success of diffusion–probabilistic models that outperform GANs on image quality benchmarks [2], the field has advanced at a remarkable pace. State-of-the-art language and multimodal models such as ChatGPT and DeepSeek now scale to hundreds of billions of parameters, enabling few-shot reasoning and open-domain dialogue [3], [4]. Yet this rapid scaling carries steep computational costs: training a single large-language model can emit large amounts of $CO_2$ and consume a lot of electricity [5]. Even with bespoke accelerators, classical hardware struggles to accommodate the memory footprint and wall-clock time required to fit such massive networks, motivating the search for novel computing paradigms.

Quantum computing (QC) offers a complementary path forward. Landmark demonstrations of quantum supremacy suggest that, for carefully chosen circuits, near-term quantum devices already outperform the largest classical supercomputers on sampling problems [6]. Although today's hardware remains noisy, small, and error-prone, quantum algorithms tailored to this Noisy Intermediate-Scale Quantum (NISQ) regime show promise in optimisation, chemistry, and crucially probabilistic modelling [7]. In particular, parameterised quantum circuits (PQCs) can express broad families of probability distributions with far fewer tunable parameters than their classical analogues [8], making them intriguing candidates for enhancing GAI.

Over the past five years, a vibrant research sub-field, Quantum Generative AI (QGAI), has emerged at this intersection. Figure 1 shows the current landscape of QGAI. Canonical examples include Quantum Circuit Born Machines (QCBMs), which sample from quantum Born probabilities to

Siva Sai and Vinay Chamola are with the Department of Electrical and Electronics Engineering, Birla Institute of Technology and Science, Pilani, Pilani Campus, Vidya Vihar, Pilani, Rajasthan 333031, India (e-mails: {p20220063, vinay.chamola}@pilani.bits-pilani.ac.in). Vinay Chamola is also with APPCAIR.

Ishika Goyal is with Department of Computer Science and Engineering, Birla Institute of Technology and Science, Pilani, Pilani Campus, Vidya Vihar, Pilani, Rajasthan 333031, India (e-mail: f20220549@pilani.bits-pilani.ac.in).

Rajkumar Buyya is with the Quantum Cloud Computing and Distributed Systems (qCLOUDS) Laboratory, School of Computing and Information Systems, The University of Melbourne, Australia (e-mail: rbuyya@unimelb.edu.au). Siva Sai is also with qCLOUDS Laboratory.

approximate classical data distributions with provably high expressive power [8]; Quantum GANs (QGANs), whose generators and discriminators may each be quantum, classical, or hybrid [9]; and Quantum Boltzmann Machines (QBMs), which leverage transverse-field Hamiltonians to model Gibbs distributions more richly than classical Restricted BMs [10]. More recent work extends diffusion–probabilistic modelling to quantum states [11], [12].

Despite encouraging progress, formidable challenges remain. Barren-plateau phenomena, where gradients vanish exponentially with circuit width or depth, can stall optimisation long before convergence [13]. Data-embedding overhead and shot noise limit throughput in hybrid workflows, while qubit decoherence imposes strict depth budgets on variational circuits. Moreover, integrating quantum accelerators with classical pre- and post-processing pipelines raises architectural, scheduling, and energy-efficiency questions that are still largely unresolved [7]. A systematic treatment of these issues is essential as the community transitions from proof-of-concept demonstrations to real-world deployments in drug discovery, finance, materials science, and human–machine interaction. GAI has made huge leaps in recent years, with models now boasting billions of parameters and being trained on datasets that seem to grow larger by the day. But as these models get bigger and more complex, we're starting to hit real walls with classical computing. The more we scale up, the more we run into issues like running out of memory, waiting forever for models to train, and watching energy costs skyrocket. This is especially true for the types of problems at the heart of modern generative models, such as GANs, VAEs, and diffusion models, which all have to juggle high-dimensional data, tricky optimization landscapes, and lots of probabilistic sampling [13], [14]. Even with all the progress in classical hardware, it's tough to capture the kinds of non-local correlations and subtle structures that make generative models really shine [13], [15]. In practice, this means we often have to compromise on either the quality or the speed of our models. Table III summarizes the major differences between classical and quantum GAI.

That's where quantum computing (QC) comes in as a game-changer. Quantum computers work in a totally different way from classical ones. Instead of bits that are either 0 or 1, quantum bits (qubits) can be in a superposition of both, and they can get entangled with each other in ways that classical bits just can't. This lets quantum systems represent and process high-dimensional probability distributions much more efficiently, with resources that grow only polynomially (or even less) as you add more qubits [16]. For generative modeling, this is a big deal because it means quantum computers can explore huge solution spaces and pick up on correlations that classical computers would struggle to even represent [13]. There are already quantum algorithms out there like amplitude estimation, QGANs, and variational circuits that show real promise in speeding up sampling, optimization, and inference tasks that are otherwise painfully slow on classical machines [17].

The benefits of quantum computing for GAI go beyond speed. Quantum models can use entanglement to create richer, more expressive latent spaces, which helps generate outputs that aren't just diverse but also more coherent and contextually relevant [15]. Techniques like quantum neural networks, quantum Boltzmann machines, and quantum natural gradients are starting to tackle some of the classic problems in deep learning, like vanishing gradients and getting stuck in local minima. This means models can learn more stably and generalize better, which is something everyone in the field is after [13], [15]. Plus, quantum-inspired algorithms are already making their way into classical generative models, hinting at what we might see as quantum hardware keeps getting better [14]. Another important advantage is energy efficiency. Training these massive generative models takes a ton of power, which is becoming a real concern as we think about scaling up and keeping things sustainable . Quantum systems, with their parallelism and the ability to run certain computations with much less depth, could offer a much more energy-efficient way to do the same work . This matters a lot for fields like healthcare, finance, and scientific research, where big generative models could make a huge impact but are often held back by the sheer computational and energy costs [13], [15].

This rest of the paper is organized as follows: Section III presents preliminaries on Generative Artificial Intelligence and Quantum Computing, providing the necessary foundations. Section IV surveys quantum-enhanced generative architectures, including Quantum Circuit Born Machines, Quantum Generative Adversarial Networks, Quantum Boltzmann Machines, Quantum Variational Autoencoders, and Quantum Diffusion Models. Section V highlights applications of QGAI across diverse domains such as human–machine interaction, drug discovery, IoT security, resource allocation, and finance. Section VI discusses key challenges and future directions, covering scalability, error mitigation, training stability, data encoding, energy efficiency, interpretability, hybrid integration, benchmarking, and algorithmic innovations. Finally, Section VIII concludes the paper by summarising the main insights and emphasising the future potential of QGAI in advancing next-generation intelligent systems.

## II. Related Work

Prior literature spans broad QAI overviews, reliability-centric analyses, AI-for-quantum systems papers, and introductory book chapters. In contrast to these, our survey concentrates specifically on quantum generative architectures (QCBM, qGAN, QBM, QVAE, quantum diffusion), their applications, and a consolidated agenda for QGAI. Table I shows unique contribution of this paper within the existing literature by comparing scope, depth, and emphasis across five representative works [18]–[22].

The SSRN overview on quantum-augmented generative modeling [22] covers principles and sketches applications, but does not provide a unified taxonomy or standardized evaluation protocols, hence the partial marks. The recent arXiv survey on "AI for Quantum" [19] concentrates on using classical ML for compilation, routing, and architecture search

Table I: Comparison of related surveys

| Ref. | Primary Focus | Key Contributions | Limitations | Princ. | Tax. | Appns. | Bench. |
|---|---|---|---|---|---|---|---|
| [19] | AI for Quantum | ML/RL for compilation, qubit routing and architecture search; notes generative approaches for circuit generation | Not a QGAI survey; focus is AI for QC rather than QGAI; minimal application taxonomy | ○ | × | ○ | × |
| [21] | QAI | Bidirectional view (QC for AI & AI for QC); enumerates data/algorithmic/framework gaps | Broad and introductory; generative models not treated in depth; limited taxonomy and evaluation protocols | ○ | × | ○ | × |
| [22] | QC–AI book chapter | Introductory exposition of QC concepts and qualitative applications/ethics; QML overview | Descriptive style; lacks technical taxonomy, benchmarks, and QGAI specificity | ○ | × | ○ | × |
| [20] | Reliable AI & computability | Theoretical analysis of reliability and computability limits on digital/quantum/analog models | No QGAI focus; lacks applications and taxonomy for generative models | ✓ | × | × | × |
| [18] | QGAI overview | Hybrid frameworks, quantum generative models, feature-space augmentation; application sketches in drug discovery and finance | Narrative review; lacks deep taxonomy and standardized benchmarks; limited implementation details | ✓ | ○ | ✓ | × |
| **Our paper** | Taxonomy with critical review and future directions | Systematic coverage of QGAI: QCBM, qGAN, QBM, QVAE, quantum diffusion; cross-domain applications; synthesized challenges and research agenda | Empirical case studies and deployments are out of scope | ✓ | ✓ | ✓ | ✓ |

**Legend:** *Princ.* = Principles; *Tax.* = Taxonomy; *Appns.* = Applications; *Bench.* = Benchmarks. *Applications* refers to whether the work surveys concrete domains (e.g., drug discovery, HMI, IoT security, finance, materials) with technical detail; *Benchmarks* refers to presence of standardized datasets/metrics, reproducible protocols, or quantitative comparisons. **Symbols:** ✓ = covered; × = not covered; ○ = partially covered (mentioned briefly or treated at a high level without comprehensive taxonomy/benchmarks).

and therefore aligns only partially with QGAI applications. The reliability-focused analysis [18] contributes theoretical grounding for trustworthy AI across computing paradigms but does not engage with generative architectures or application taxonomies. The book chapter on the QC–AI interplay [20] provides a broad introduction and problem landscape without a dedicated generative taxonomy or experimental case studies. The introductory chapter on "Quantum Leap" [21] surveys concepts and qualitative applications, again without a detailed taxonomic structure or benchmarking.

## III. Background

This section provides a foundational overview of the two key pillars that underpin the Quantum Generative AI landscape: Generative Artificial Intelligence and Quantum Computing. Understanding their fundamental concepts, capabilities, and limitations is essential before diving into QGAI-specific models, architectures, and applications. We begin with GAI, a field that has revolutionized data synthesis and generative learning, followed by an introduction to the principles and computational advantages of quantum computation.

### A. Generative Artificial Intelligence

Generative artificial intelligence (GAI) refers to a class of powerful models that learned to produce realistic and novel data, such as images, text, and audio. The rapid growth of GAI was driven by advancements in deep learning, massive datasets, and increased computational power, leading to widespread applications across various fields, including creative arts, scientific research, and commerce. This section explored the core principles of GAI, its primary model families, and the key limitations that continue to challenge its development and deployment.

*Probabilistic Foundations:* At the heart of GAI lies the goal of modeling a data-generating distribution $q(\mathbf{x})$ over a high-dimensional data space $\mathcal{X}$, such that a model can generate novel, high-quality samples $\tilde{\mathbf{x}} \sim p_\theta(\mathbf{x})$ that resemble real data . To accomplish this, GAI systems typically define a parameterized distribution $p_\theta(\mathbf{x})$ and train it by minimizing some statistical divergence,such as the Kullback–Leibler (KL) divergence, Jensen–Shannon (JS) divergence, or Wasserstein distance,between the real and model distributions. This is generally done through stochastic gradient descent, often using surrogate loss functions suited to the model class [23].

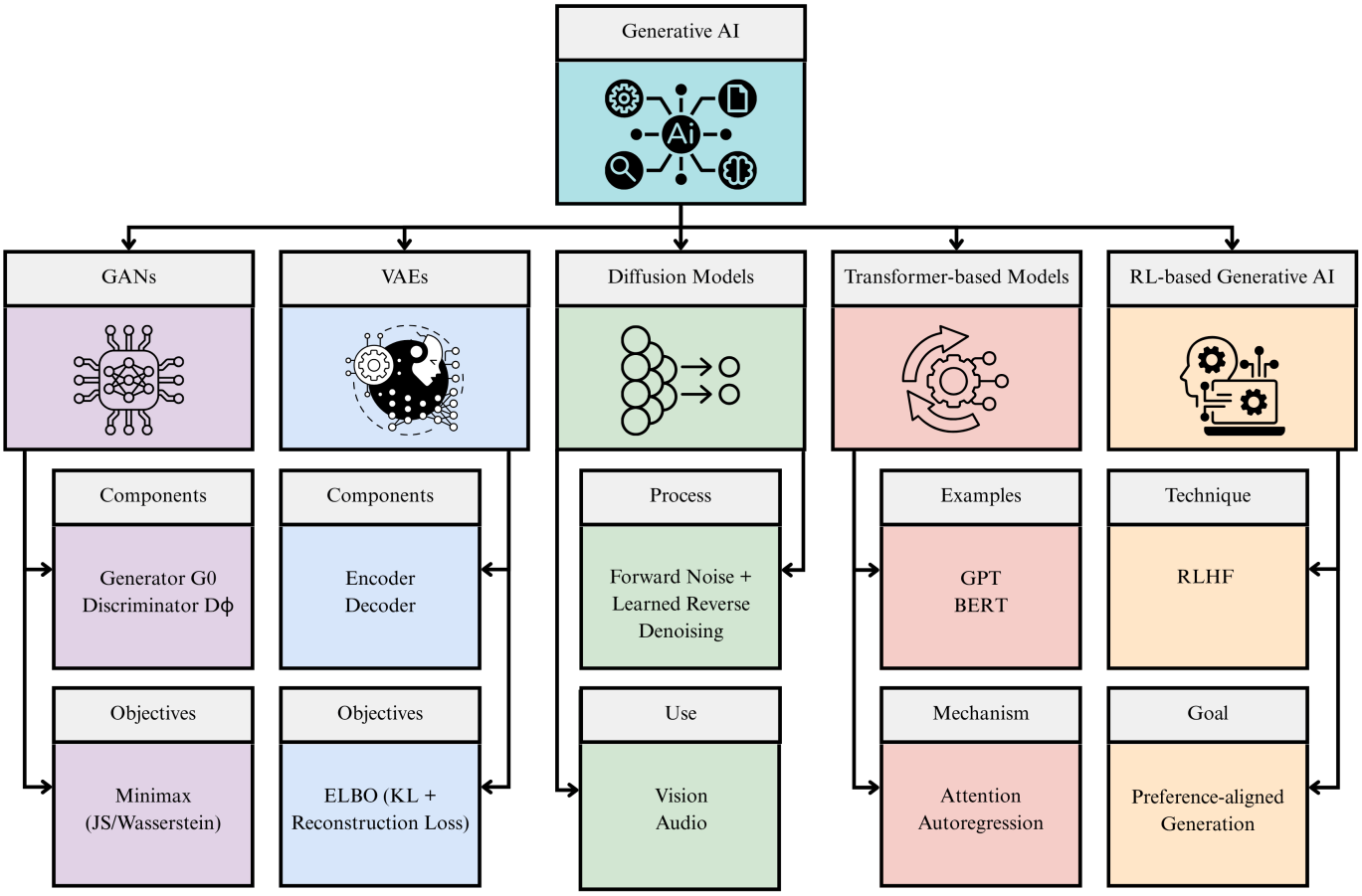


Figure 2: Taxonomy of Generative AI Models

*Model Families:* Several generative model paradigms have emerged as dominant approaches, as summarized in Figure 2. Generative adversarial networks (GANs) train a generator $G_\theta$ and discriminator $D_\phi$ in a two-player minimax game, reaching equilibrium when real and synthetic samples are indistinguishable [23]. Variational autoencoders (VAEs) use probabilistic encoders and decoders to optimize an evidence lower bound, balancing reconstruction accuracy with latent-space regularization [24]. Diffusion models simulate a forward noising process and a learned reverse denoising process, often yielding state-of-the-art sample quality in vision and audio [2]. Transformer-based models (e.g., GPT) employ attention mechanisms to model sequences autoregressively or bidirectionally, enabling strong results in language, code, and captioning [25]. Finally, reinforcement learning for generative AI, including RLHF, aligns model outputs with preferences or downstream utility through policy optimization [26].

*Classical Limitations:* Despite the success of generative AI, three constraints persist. First, training cost remains high: large models (for example, GPT-3) are trained on trillion-token corpora and require substantial compute and energy. Second, sampling latency is significant because transformer and diffusion families are sequential or iterative at inference. Third, expressivity is limited for entangled or highly non-local dependencies that arise in quantum physics, genomics, and financial systems [27]. These constraints motivate quantum computing as a candidate for next-generation generative systems.

### B. Quantum Computing (QC)

Quantum computing leverages quantum-mechanical principles to perform computation. Unlike classical bits, qubits can exist in superpositions of 0 and 1, and entanglement allows correlations that do not admit classical descriptions, enabling potential advantages on specific problem classes.

*Fundamentals of Quantum Information:* Quantum information is carried by qubits, which are two-level systems represented as $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$. A register of $n$ qubits inhabits a $2^n$-dimensional Hilbert space, providing exponential representational capacity. Computation proceeds via unitary gates that evolve the state reversibly, and measurement projects the state to classical outcomes according to the Born rule.

*Quantum Algorithms:* Several algorithms illustrate potential advantages. Shor's algorithm factors large integers in polynomial time, challenging RSA security and delivering an exponential speedup over the best known classical methods [28]. Grover's algorithm searches unstructured databases in $O(\sqrt{N})$ queries, yielding a quadratic speedup [29]. Variational quantum algorithms (VQAs) target the NISQ regime by combining parameterised quantum circuits with classical optimisers; prominent examples include the Variational

Quantum Eigensolver (VQE) and the Quantum Approximate Optimisation Algorithm (QAOA) [7].

*Hardware Capabilities and Constraints:* Leading architectures for QC include superconducting qubits (IBM, Google) [30], trapped ions (IonQ) [31], photonic circuits (Xanadu) [32], and neutral atoms (QuEra) [33]. Practical systems, however, face limitations in qubit count and quality, with error rates above fault-tolerant thresholds; coherence times and gate fidelities restrict feasible circuit depth; and the classical–quantum interface introduces latency that necessitates efficient co-processing strategies.

*Why QC for GAI?:* Two principles are especially relevant for generative modelling. Exponential encoding allows an $n$-qubit state to represent a distribution over $2^n$ outcomes, supporting compact modelling of complex manifolds [34]. In addition, sampling arises natively: quantum circuits emit outcomes according to their internal probability amplitudes, avoiding explicit softmax layers or temperature tuning and enabling direct generation from the modelled distribution [35].

Table II presents a comparative overview of key Quantum Generative Models (QGMs). It highlights the differences in their architectures, training objective functions, and the nature of their generated outputs. This provides a structured understanding of how various QGMs approach generative learning in quantum settings.

## IV. Architectures

Quantum-enhanced generative AI (GAI) architectures aim to leverage quantum computing principles to improve model efficiency, scalability, and representational power (see Figure 3).

### A. Quantum Circuit Born Machines

Quantum Circuit Born Machines (QCBMs) are unsupervised quantum generative models that leverage parametrized quantum circuits (PQCs) to approximate the probability distribution of a classical dataset. By producing tunable discrete distributions $p_\theta$, a QCBM seeks to match a given target distribution $q$, and has proven effective on both synthetic and real-world data. The key advantage is the high expressivity of quantum circuits, which allows them to capture complex correlations that are difficult for classical models to represent [8]. Figure 4 represents the training mechanism of QCBMs.

A typical QCBM consists of alternating layers of single-qubit rotation gates (e.g., $R_X$, $R_Y$, $R_Z$) and multi-qubit entangling gates (e.g., CNOT, CZ). Shallow circuits with local connectivity can model simple distributions, whereas deeper, all-to-all entanglement can encode far richer correlations. Training proceeds by minimizing a divergence metric, often Kullback–Leibler (KL) divergence or maximum mean discrepancy (MMD), between samples from $p_\theta$ and the target distribution. Gradients (or gradient-free updates) are used by optimizers such as Adam, CMA-ES, SPSA, or particle-swarm optimization, with the choice dictated by noise tolerance and problem scale [35]. Benedetti et al. [8] first formalized the QCBM framework and demonstrated it on trapped-ion hardware, establishing early training protocols. Zhu et al. [36] then showed that QCBMs remain viable on a seven-qubit trapped-ion device, validating scalability under realistic noise. Liu and Wang [37] introduced gradient-based optimizers, achieving faster convergence.

Table II: Comparison of key Quantum Generative Models.

| **Model** | **Architecture** | **Objective Function** | **Target Output** |
|---|---|---|---|
| Quantum GAN (QGAN) | PQC + Classical Discriminator | Minimax (Wasserstein / JS) | Classical or Quantum Data |
| Quantum VAE (QVAE) | Quantum Encoder + Decoder | ELBO (Fidelity + Regulari -zation) | Quantum State or Distribution |
| Quantum Diffusion (QDM) | PQC with Temporal Unfolding | Fidelity Loss (Reverse Process) | Noisy or Pure States |
| Quantum Circuit Born Machine (QCBM) | PQC Only | MMD or KL Divergence | Discrete Distribution |

Optimization strategies have also diversified. Particle-swarm optimization and CMA-ES showed early promise on noisy devices [38], while Bayesian optimization excelled on high-parameter landscapes [36]. Riofrío et al. [39] provided a systematic comparison of QCBMs and QGANs, showing that shallow QCBMs struggle with complex correlations whereas deeper circuits improve fidelity at the cost of optimization difficulty,highlighting a fundamental depth–trainability trade-off. Finally, Zhou et al. [40] used a matrix-product-state (MPS) architecture to compress QCBMs for NISQ hardware. Their MPS-QCBM reproduced 2 × 2 Bars-and-Stripes data using only two or three physical qubits (with bond-dimension "virtual" qubits), matching conventional QCBM accuracy while reducing hardware requirements,at the expense of a larger parameter space that can slow optimization for bigger datasets.

### B. Quantum Generative Adversarial Networks

Quantum Generative Adversarial Networks (QGANs), first proposed by Lloyd and Weedbrook [9], transpose the classical GAN paradigm into the quantum domain. A parametrised quantum circuit (PQC) plays the role of the generator, emitting quantum states whose measurement outcomes define a synthetic data distribution, while a discriminator implemented either classically or quantum mechanically learns to tell these samples from those drawn from the target distribution. Training proceeds as a minimax optimisation: the generator minimises, and the discriminator maximises, a distance between the two

Table III: Classical vs Quantum Generative AI

| **Aspect** | **Classical GAI** | **Quantum GAI** |
|---|---|---|
| Training cost | Large models (e.g., GPT-3) require trillion-token datasets, high compute and energy | Parameterized quantum circuits can reduce sample and compute requirements |
| Sampling latency | Transformer and diffusion models incur large inference delays | Quantum circuits output samples directly via the Born rule |
| Expressivity | Limited in encoding entangled or highly non-local dependencies | Exponential encoding and entanglement capture complex correlations |
| Hardware constraints | Memory, wall-clock time, $CO_2$ emissions | NISQ devices: noisy, small qubit counts, decoherence constraints |

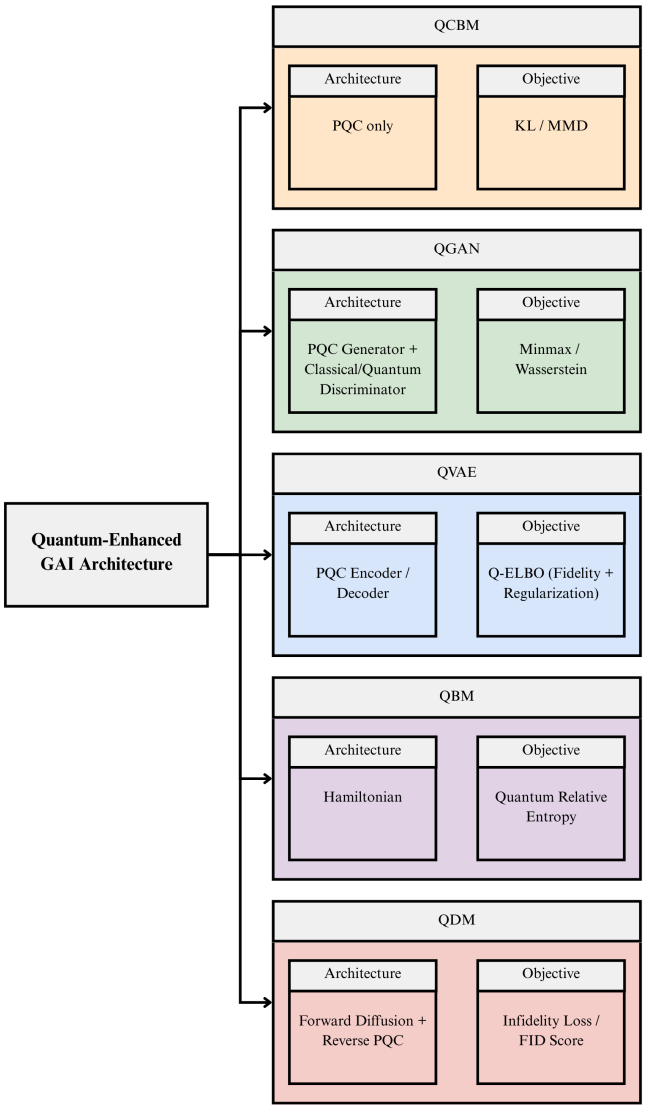


Figure 3: Taxonomy of Quantum Generative Architectures

distributions. Because the generator lives in Hilbert space, QGAN training can exploit quantum-native routines such as the variational-quantum-eigensolver, enabling a more efficient search of high-dimensional probability landscapes than is possible with purely classical stochastic-gradient updates. Figure 5 shows the schematic working of QGANs.

Lloyd and Weedbrook outlined four architectural flavours, determined by whether the task data and the two agents are classical (C) or quantum (Q): QT-QGQD (quantum task, quantum generator, quantum discriminator), QT-CGQD, QT-CGCD, and CT-QGCD [9]. The fully quantum option offers the greatest expressive power but is the most hardware intensive; the hybrid variants trade quantum advantage for near-term feasibility.

Riofrío et al. [39] systematically benchmarked QGANs against Quantum Circuit Born Machines (QCBMs), showing that QGAN generators can approximate classically intractable distributions yet suffer more severely from mode collapse and training instabilities on noisy hardware. To mitigate those issues, several design improvements have been proposed:

(1) Tensor-network generators – Huggins et al. [41] replaced the PQC with matrix-product or tree tensor networks, enabling classical pre-training before transfer to quantum hardware and reducing circuit depth.

(2) Patch-based QGANs – Huang et al. [42] introduced a quantum patch-GAN that divides data into smaller patches, each handled by a shallow sub-generator, making image generation possible with limited qubits.

(3) Quantum Wasserstein losses – Chakrabarti et al. [43] adapted the Wasserstein distance to quantum settings to stabilize training. Kiani et al. [44] refined the metric to remove unitary-invariance artifacts and boost state fidelity.

(4) Efficient gradients – The parameter-shift rule, generalised to noisy hardware by Schuld et al. [45], provides unbiased gradient estimates without ancillary qubits and is now standard for PQC optimisation.

### *C. Quantum Boltzmann Machines*

Quantum Boltzmann Machines (QBMs) generalise classical Boltzmann machines by replacing Ising spins with qubits governed by a transverse-field Hamiltonian. First proposed by Amin et al. [10], a QBM defines a quantum Boltzmann distribution whose Gibbs state can encode richer correlations than any classical counterpart. Visible qubits represent data,

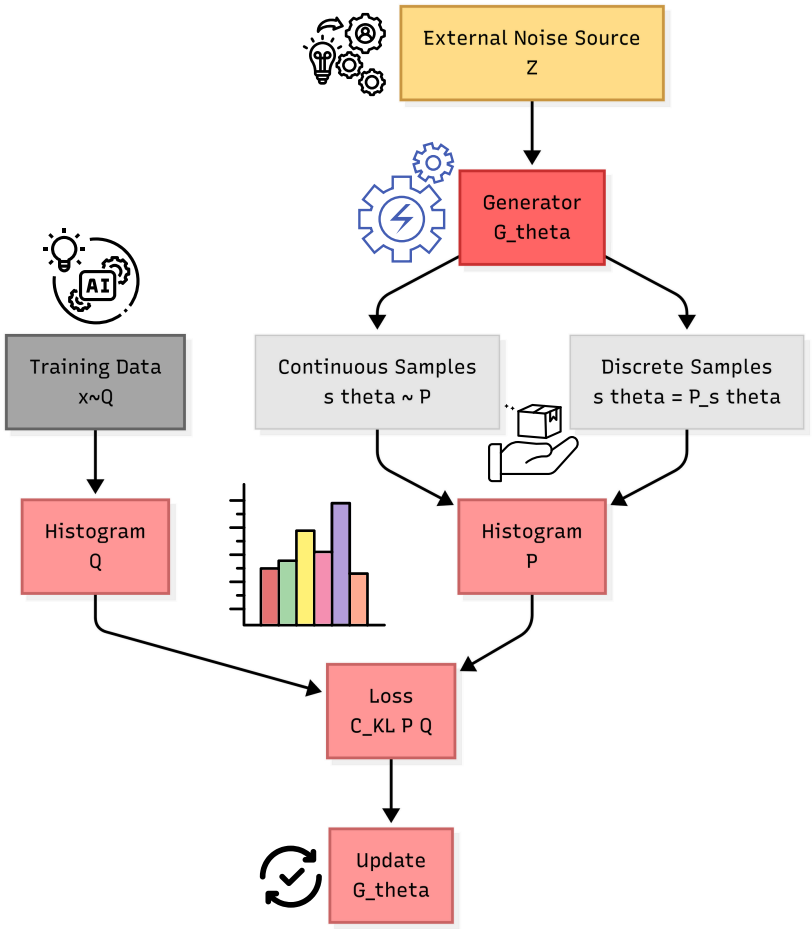


Figure 4: Schematic of training of QCBM

hidden qubits supply latent capacity, and trainable couplings appear as longitudinal and transverse fields. Learning proceeds by minimising a distance, often quantum relative entropy between model and data distributions, but non-commuting Hamiltonian terms make gradients expensive. Practical training, therefore, uses approximations such as bound-based objectives or sampling on quantum annealers, where QBMs enjoy a natural hardware realisation.

Different types of QBMs are listed as follows:

*a) Semi-restricted QBMs.:* Amin et al. [10] showed that forbidding hidden–hidden couplings simplifies training while retaining quantum advantage. With visible units clamped during optimization, the semi-restricted QBM converged faster than fully connected RBMs and captured higher-order correlations.

*b) State-based training for tomography:* Kieferová and Wiebe [46] introduced state-based training, where a QBM is fitted directly to measurement data from an unknown quantum state—circumventing full-state comparison and enabling scalable quantum-state tomography on near-term devices.

*c) Reinforcement-learning:* Crawford et al. [47] embedded QBMs in policy-evaluation loops, demonstrating higher success rates than classical RBMs on standard reinforcement-learning benchmarks by exploiting quantum-induced correlations.

*d) Variational QBMs:* Zoufal et al. [48] re-cast the QBM as a parametrised quantum circuit that prepares Gibbs states variationally. Exact gradients obtained via the parameter-shift rule reduce sampling overhead and improve scalability to larger lattices—useful, for instance, in quantum-chemistry Hamiltonians.

*e) Evolved QBMs:* Minervini et al. [49] proposed evolved QBMs (EQBMs), which apply a second Hamiltonian $H(\phi)$ to evolve the thermal state of a base Hamiltonian $G(\theta)$. The resulting state $\omega(\theta, \phi) = e^{-iH(\phi)}\rho(\theta)e^{iH(\phi)}$ combines imaginary and real-time dynamics, enlarging the model family while still admitting analytic gradients and natural-gradient updates driven by quantum Fisher information.

*f) Sample-efficient training:* Coopmans and Benedetti [50] reframed QBM learning as matching expectation values of Hamiltonian terms, proving that stochastic gradient descent converges in polynomial time and samples when those expectations are estimated with shadow tomography. Pre-training via mean-field or Gaussian-Fermionic models further mitigates barren plateaus.

### D. Quantum Variational Autoencoders

Quantum Variational Autoencoders (QVAEs), first proposed by Khoshaman et al. [51], extend the classical

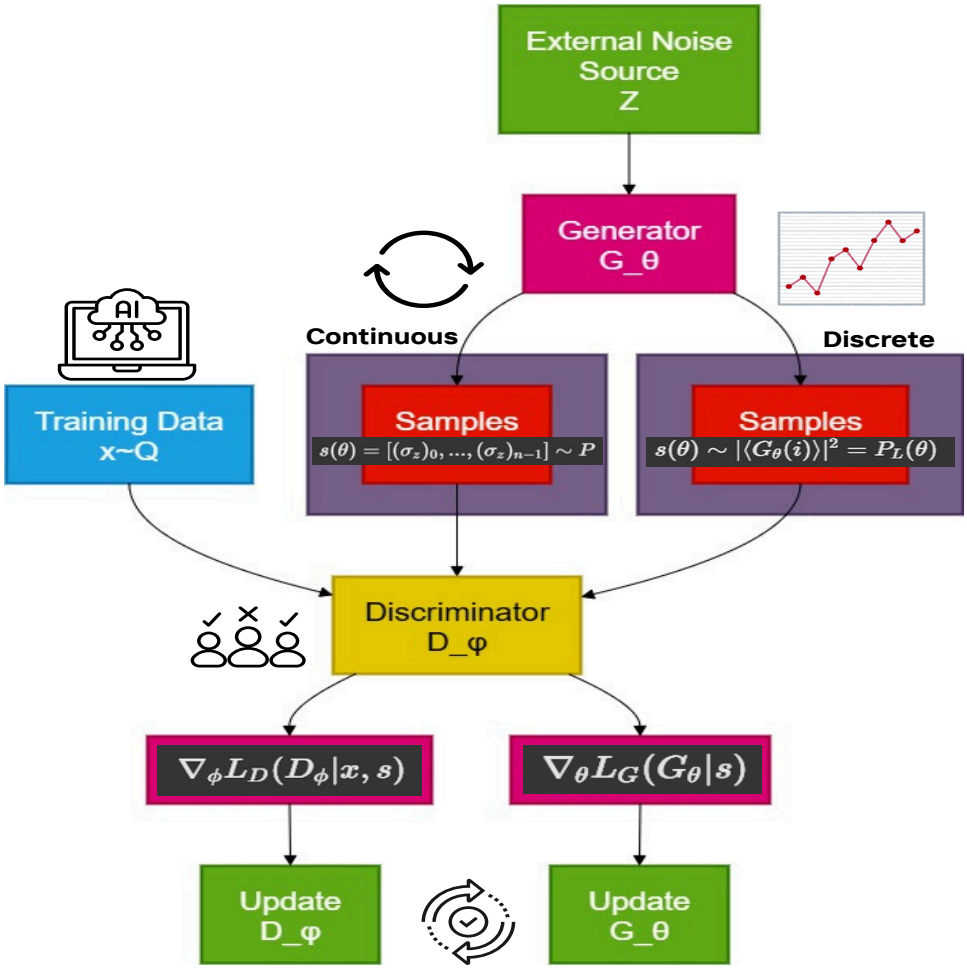


Figure 5: Schematic of training of QGAN

variational-autoencoder paradigm by inserting parametrised quantum circuits (PQCs) into the encoder and decoder. The quantum encoder maps classical or quantum input data into a low-dimensional latent space, and the decoder reconstructs the original input from this compressed representation. Training minimises a quantum evidence lower bound (Q-ELBO) that balances reconstruction fidelity (via, e.g., state-fidelity metrics) against latent-space regularisation to a chosen prior. By sidestepping a full comparison with exponentially large target states, QVAEs are well suited to near-term hardware: the PQCs provide compact dimensionality reduction while preserving essential data features [51].

Figure 6 shows QVAE architecture along with classical VAE architecture. The classical VAE architecture includes a decoder and an encoder which outputs Gaussian parameters for approximating the latent space distribution, from which random noises are sampled for generating new data through the decoder network (the generation or sampling process is denoted in red box and arrows); (b) a patched variational quantum circuit (each patch has three qubits) serves as the quantum encoder/decoder network with corresponding quantum input, hidden, and output layers. A hybrid VAE/AE is realized by connecting the final fully connected (FC) layer with the preceding measurement outcome vector.

Different types of QVAEs are listed as follows:

*a) EF-QAE:* Bravo-Prieto et al. [52] introduced the Enhanced-Feature Quantum Autoencoder (EF-QAE), which replaces global cost functions with local ones to avoid vanishing gradients during training . Encoding each data component into a single-qubit rotation and optimizing local fidelities yielded high-accuracy compression of both quantum many-body states and classical images.

*b) Hybrid QVAE:* Srikumar et al. [53] introduced a hybrid quantum autoencoder (HQA) that extracts information from quantum states by mapping them into a classical representational space. The approach enables clustering and semi-supervised classification of pure states, demonstrated on amplitude-encoded states, with potential extensions to more complex quantum data.

*c) Noise-assisted and adiabatic QAEs:* Cao and Wang [54] introduced a noise-assisted quantum autoencoder that improves recovery fidelity by aligning input and output mixedness using noise channels determined from measurement data. The approach utilizes measurement information more effectively, extends to both circuit and adiabatic models (implementable on quantum annealers), and is validated on thermal states of the transverse-field Ising model and Werner states. A projected quantum autoencoder is also proposed for

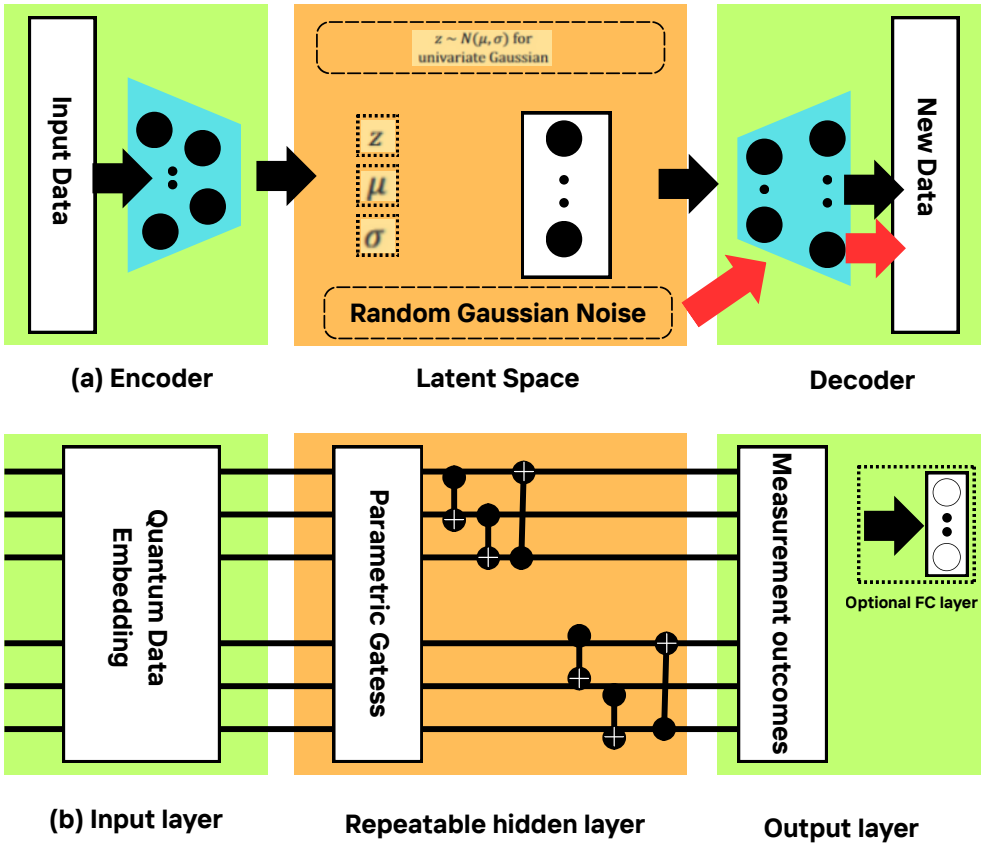


Figure 6: QVAE architecture

pure-state ensemble compression.

*d) ζ-QVAE:* Quantum computing faces challenges in handling large real-world datasets due to limited hardware resources. To address this, the authors in [55] proposed ζ-QVAE, a fully quantum variational autoencoder that compresses both classical and quantum data into low-dimensional latent representations while enabling effective reconstruction. Unlike prior quantum compression models, ζ-QVAE employs regularized mixed-state latent spaces, supports multiple divergences, and leverages full density matrices for efficient optimization. This framework mirrors the capabilities of classical VAEs while remaining fully quantum, with applications demonstrated on genomics and synthetic data, showing improved latent space utilization and competitive or superior performance compared to classical models.

### *E. Quantum Diffusion Models*

Quantum diffusion models (QDMs) translate classical diffusion–probabilistic frameworks into the quantum domain by replacing neural networks with parametrised quantum circuits (PQCs). Cacioppo et al. [56] introduced the first such model: a classical forward process adds Gaussian noise, while a quantum backward process iteratively denoises via a PQC, eliminating repeated state-encoding overhead and keeping the workflow NISQ-friendly. Amplitude encoding maps classical data (e.g., images) into quantum states, and the PQC then minimizes an infidelity loss between denoised and target states.

Following are the different variants of QDMs:

*a) Latent and fully quantum QDMs.:* Cacioppo et al. [56] explored both a fully quantum generator and a latent variant that first compresses data with a classical autoencoder before quantum denoising. The latent model runs on today's NISQ hardware and supports conditional generation by encoding labels in auxiliary qubits. MNIST experiments produced recognizable digits, though the current Fréchet Inception Distance (FID) still lags behind classical diffusion models.

*b) QuDDPM.:* Zhang et al. [11] proposed Quantum Denoising Diffusion Probabilistic Models, using quantum scrambling in the forward direction and a PQC-based reverse process trained with hybrid (classical and quantum) gradients. They achieved $>99\%$ state-generation fidelity for eight-qubit mixed states and reported polynomial sample complexity, outperforming QGAN baselines.

*c) QGDM and RQGDM:* Chen et al. [12] introduced the Quantum Generative Diffusion Model, whose non-unitary forward step uses depolarising channels to reach a maximally mixed state. A trainable, timestep-embedded PQC then reconstructs the target, yielding 99.9% fidelity on pure-state tasks and 99.3% on eight-qubit mixed states, again surpassing QGANs. A resource-efficient variant (RQGDM) reduces auxiliary-qubit demands without sacrificing accuracy.

*d) Q-Dense and QU-Net:* Kölle et al. [57] introduced two quantum diffusion architectures, Q-Dense and QU-Net, designed to enhance image generation with quantum machine

learning. Q-Dense employs amplitude embedding within dense quantum circuits, while QU-Net serves as a quantum analogue of the classical U-Net, incorporating quantum convolutional layers. To accelerate generation, the authors proposed a unitary single-sampling circuit that compresses the traditional multi-step reverse diffusion process into a single operation. Benchmarking on datasets such as MNIST, Fashion MNIST, and CIFAR-10 showed that these quantum models achieved superior performance compared to classical counterparts with a similar number of parameters, as measured by metrics like FID, SSIM, and PSNR. These results demonstrate the potential of quantum diffusion models to address efficiency and parameter-scaling challenges in generative modeling.

Figure 7 illustrates the QU-Net architecture that integrates quantum convolutions. The model leverages down blocks, up blocks, and skip connections to process input slices, embedding them into a dense quantum circuit. This design enables hierarchical feature extraction and efficient reconstruction, akin to a quantum analogue of the classical U-Net. Table IV summarizes all the above-mentioned quantum architectures.

## V. Applications

The integration of quantum computing and generative AI represented a significant advancement, establishing a new frontier for addressing problems that were previously intractable for classical systems. This emergent field, referred to as QGAI, leveraged the fundamental principles of quantum mechanics, such as superposition and entanglement, to model and generate complex data with unprecedented efficiency. QGAI demonstrated its potential across a range of disciplines, from enhancing human–machine interaction to accelerating the discovery of new therapeutic compounds. This section reviews some of the most significant applications of QGAI, illustrating how these innovative models began to address some of the most formidable challenges in science and technology.

### A. Human–Machine Interaction

Human–machine interaction (HMI) studies intuitive interfaces and collaborative workflows that let people and intelligent systems communicate, learn, and execute tasks seamlessly [58]. Classical AI, especially natural-language processing (NLP) and decision-making, underpins most current HMI frameworks, yet struggles with high-dimensional data, real-time adaptability, and cross-cultural language barriers [59]. For example, NLP engines often misinterpret dialectal variations or cultural nuances, reducing effectiveness in multilingual or rapidly changing settings [60]. Adaptive HMI must also balance clarity with complexity: poorly designed interfaces can overload users or cause miscommunication during shared tasks [61].

Quantum Generative AI (QGAI) can alleviate these issues by combining techniques like superposition and entanglement with powerful generative models. Quantum circuits process multimodal data (text, speech, images) in parallel, enabling QGAI systems to learn user preferences and close feedback loops in real time [22]. QGAI also mitigates NLP shortcomings by generating culturally nuanced translations and disambiguating intent through quantum-optimised context analysis [62]. These advances make QGAI a promising foundation for next-generation HMI that demands robust cross-cultural interoperability and real-time adaptivity. Lee et al. [63] introduced a Quantum Computational Intelligence (QCI) agent with a content attention ontology model to improve human–machine interaction in Taiwanese/English language learning. It integrates a Generative AI image agent, SBERT(Sentence Bidirectional Encoder Representations from Transformers)-based similarity scoring, and Meta AI's Universal Speech Translator to support bilingual text–image generation and dialogue experiences. Using Particle Swarm Optimization (PSO), the system predicts learner performance, while the enhanced NUTN.TW-LLM enables interactive and adaptive learning. Experimental results showed the model's efficacy, with accuracy for social media data analysis improving to 71% and a specific language learning model achieving 90% accuracy. Human inspectors also gave the GAI image agent high grades for its generated images. Overall, the findings suggested that students were interested in co-learning languages and QCI with machines.

### B. Drug Discovery

Drug discovery seeks and optimises compounds for therapeutic use, typically via virtual screening, molecular-dynamics simulations, and laboratory validation [64], [65]. The search space, however, overwhelms classical pipelines, which are expensive and sometimes inaccurate at quantum-level phenomena. Although classical machine learning accelerates screening, it can falter when modelling high-dimensional molecular systems or predicting electronic structures and reaction pathways [66].

Quantum Generative AI tackles these hurdles by encoding molecules (graphs or SMILES) directly into quantum states. QGANs and Quantum Variational Autoencoders (QVAEs) explore chemical space via parametrised quantum circuits, efficiently sampling regions that would be classically intractable [51], [67]. These models have already pushed drug-likeness metrics such as Quantitative Estimate of Drug-likeness (QED) and partition coefficient (logP) beyond classical baselines [68].

Li et al. [69] proposed a Scalable Quantum Variational Autoencoder (SQ-VAE) and its vanilla variant (SQ-AE) for drug molecule design, combining quantum and classical architectures. By applying strategies like adjustable quantum depth and patched circuits, the models improve reconstruction and sampling of drug molecules, showing advantages in low-dimensional cases and better drug properties in high-dimensional learning compared to classical counterparts. The results showed that for low-dimensional datasets, the quantum models achieved computational advantages in training speed and representation quality. For high-dimensional molecules, the quantum generative autoencoders generated new molecules with better drug properties. For example, a scalable quantum VAE with an LSD of 18 generated molecules with a logP score of 0.780,

Table IV: Summary of Quantum-enhanced GAI Architectures

| Architecture | Introduction | Variants / Key Contributions |
|---|---|---|
| Quantum Circuit Born Machines (QCBMs) | Unsupervised models using parametrized quantum circuits (PQCs) to approximate target distributions, leveraging high expressivity of quantum circuits to capture complex correlations [8], [35]. | Early work: Benedetti et al. [8], Zhu et al. [36], Liu and Wang [37].<br>Optimizers: particle-swarm, CMA-ES, Bayesian [36], [38].<br>Riofrío et al. compared QCBMs vs QGANs [39].<br>Zhou et al.: MPS-QCBM for NISQ devices [40]. |
| Quantum Generative Adversarial Networks (QGANs) | Quantum adaptation of GANs with PQC generator and classical/quantum discriminator; training as minimax optimization in Hilbert space [9]. | Lloyd & Weedbrook: four variants (QT-QGQD, QT-CGQD, QT-CGCD, CT-QGCD) [9].<br>Benchmarked vs QCBMs (Riofrío et al.) [39].<br>Improvements: tensor-network generators [41], patch-based QGANs [42], quantum Wasserstein losses [43], [44], parameter-shift gradients [45]. |
| Quantum Boltzmann Machines (QBMs) | Generalise classical BMs with qubits and transverse-field Hamiltonians; encode Gibbs states with richer correlations; training via relative entropy or annealers [10]. | Semi-restricted QBMs [10].<br>State-based training for tomography [46].<br>RL applications [47].<br>Variational QBMs [48].<br>Evolved QBMs (EQBMs) [49].<br>Sample-efficient training guarantees [50]. |
| Quantum Variational Autoencoders (QVAEs) | Extend VAEs by embedding PQCs in encoder/decoder; trained with quantum ELBO for dimensionality reduction while preserving key features [51]. | EF-QAE [52].<br>Hybrid QVAE (HQA) [53].<br>Noise-assisted/adibatic QAEs [54].<br>$\zeta$-QVAE for fully quantum latent spaces [54]. |
| Quantum Diffusion Models (QDMs) | Translate diffusion–probabilistic models into quantum domain with PQC-based denoising; enable quantum-native reverse processes [56]. | Latent vs fully quantum QDMs [56].<br>QuDDPM [11].<br>QGDM & RQGDM [12].<br>Q-Dense & QU-Net for images [57]. |

outperforming a classical VAE's 0.357. Another variant with a LSD of 56 achieved a QED score of 0.204, compared to the classical VAE's 0.139.

Jain and Ganguly [70] introduced hybrid Quantum GAN architectures (QWGAN-HG, QGAN-HG-GP, and QWGAN-HG-GP) for molecular simulation and drug discovery. These models used a hybrid generator that combined a parameterized quantum circuit with a classical neural network to explore the QM9 dataset. The models were evaluated using statistical distance metrics and various drug property scores to assess their efficacy. The results showed that the proposed QWGAN-GP-HG-P4-L2 architecture performed best, achieving a Drug Candidate Score of 0.204 and a Validity Score of 22.469, which were significantly higher than the classical MolGAN's scores of 0.118 and 0.125, respectively.

### C. IoT Security

IoT security aims to protect connected devices, networks, and data in domains such as healthcare, smart-city infrastructure, and industrial automation. These environments face unauthorised access, data breaches, and device spoofing because they are decentralised, resource-constrained, and often rely on legacy public-key protocols (e.g. RSA, ECC) that are susceptible to quantum attacks [71]. Classical counter-measures struggle to scale, and quantum computers threaten to break many standard cryptosystems by accelerating integer factorisation and related problems [71].

In [72], the authors proposed an unsupervised anomaly detection method for high-energy physics using a quantum generative adversarial network (qGAN) trained on Standard Model (SM) data. The methodology involved training a qGAN on a limited dataset of Standard Model events and then using it to detect anomalous Higgs and Graviton events. For the quantum approach, the researchers randomly selected 100 events from each dataset, which was ten times fewer than the 1,000 events used for the classical benchmark. The experimental results indicated that the quantum model achieved comparable classification accuracy while using significantly less training data. For instance, on the 7-feature dataset, the qGAN achieved a Graviton detection AUC of 0.96 and a Higgs detection AUC of 0.92, which were comparable to or better than the classical model's respective AUCs of 0.96 and 0.87. The authors also found that the quantum model exhibited an improved expressive power, measured by a higher effective dimension.

Prajapat et al. [73] proposed a novel security protocol for AIoT-based healthcare systems by integrating Generative AI (GAI), blockchain, and quantum encryption. The protocol employs encoding, challenge–response authentication with hash/digital signatures, and dynamic key exchange to ensure

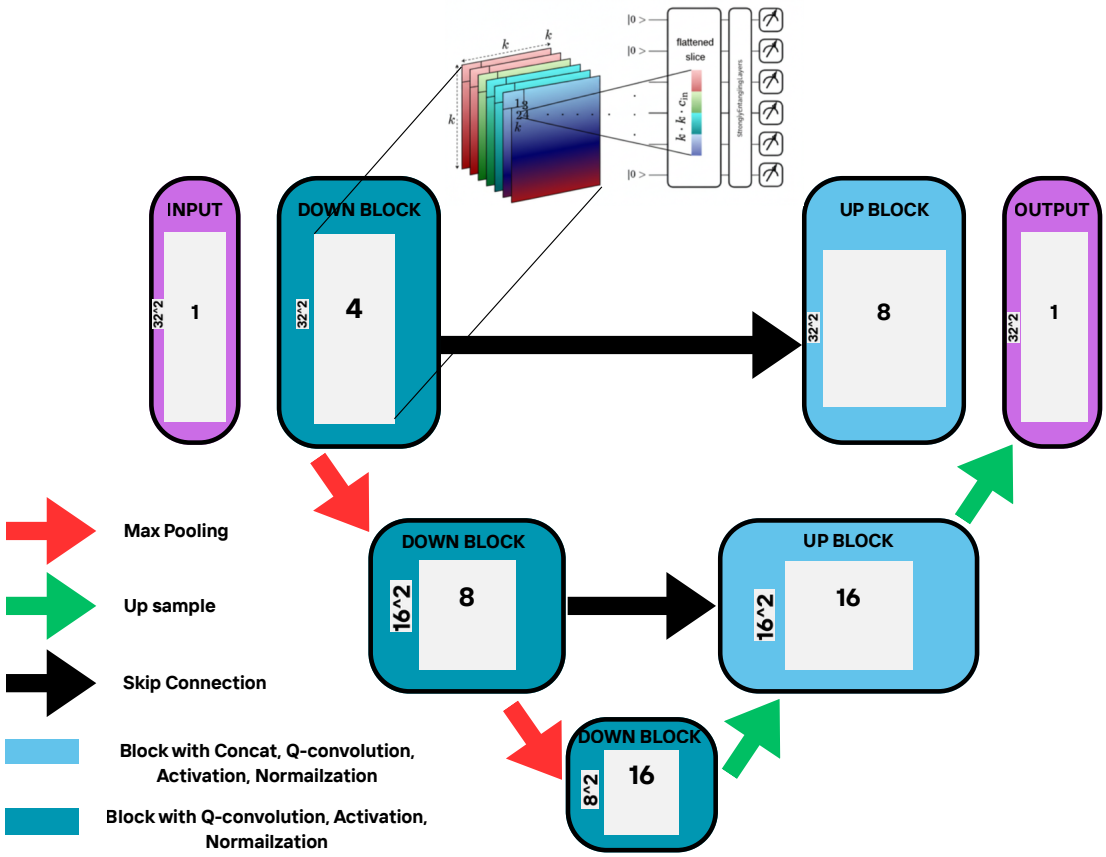


Figure 7: QU-Net architecture and quantum convolution, embedding a flattened slice into a dense quantum circuit.

secure data transmission. Experimental results show high performance with 99.4% accuracy, 99.10% precision, 98.66% recall, and 98.50% F1-score, highlighting its effectiveness in safeguarding patient data and reducing carbon-intensive inefficiencies.

### D. Resource Allocation

Quantum-computing networks must juggle qubit decoherence, fluctuating entanglement fidelity and heterogeneous service demands. Static allocation wastes resources, whereas naive dynamic policies overlook quantum-specific constraints such as variable qubit lifetimes or purification overheads. QGAI tackles the problem by marrying quantum generative models with reinforcement learning (RL). Quantum GANs and QVAEs embed high-dimensional resource states like numbers of free qubits, entangled pairs, and buffer times into parametrised circuits that predict near-optimal allocation under noise and uncertainty. They can decide, for example, when to devote extra qubits to error correction or how to route tasks across edge cloud nodes to minimise latency, all while exploiting correlations inaccessible to classical optimisers.

Xu et al. [74] proposed a multi-agent reinforcement learning (MARL) framework for decentralized resource allocation in heterogeneous quantum computing networks. By modeling each node as an agent in a partially observable environment, the method optimizes task offloading for quantum generative learning applications, such as quantum autoencoders. Experiments show that the MARL approach converges to optimal policies, reducing system costs by at least 80% compared to random allocation under various quality-of-service settings.

### E. Miscellaneous

*a) Finance.:* Quantum Circuit Born Machines (QCBMs) can model high-dimensional financial distributions such as joint asset returns and copulas with exponentially fewer parameters than classical approaches. Zhu et al. [36] simulated stock-market copulas on trapped-ion hardware, reaching 98 % fidelity for non-Gaussian tail correlations using only eight qubits, enabling real-time risk assessment for derivative pricing. Borras et al. [75] compared Quantum GANs and QCBMs on cryptocurrency data[1] (BNBBTC, ETHBTC, LTCBTC, NEOBTC, QTUMETH), equilibrium was achieved when generator and discriminator losses converged to

$$L_G = L_D = -\log\left(\tfrac{1}{2}\right) \approx 0.6931,$$

and qGANs required fewer training iterations than classical baselines, indicating a quantum advantage in efficiency. For the QCBM, COBYLA outperformed SPSA and Nelder–Mead in minimizing distribution discrepancies.

[1] https://www.tradingview.com/symbols/BNBBTC/

*b) Quantum-state compression.:* Quantum autoencoders (QAEs) mitigate hardware noise by encoding states into lower-dimensional, error-resilient subspaces. Romero et al. [76] introduced the concept of a quantum autoencoder to compress quantum data, a task for which classical compression algorithms cannot be employed. The methodology involved training a programmable quantum circuit using classical optimization algorithms to find a unitary transformation that preserved quantum information through a smaller latent space. The cost function was defined as the average fidelity between the "trash state" and a reference state, which was measured via a SWAP test in a hybrid quantum-classical scheme. The model was applied to compress the ground states of the Hubbard model and molecular Hamiltonians, including the hydrogen molecule. The results showed that both circuits tested were able to achieve high fidelities. For example, when compressing the hydrogen molecule's wavefunction from 4 to 2 qubits, Circuit A achieved a fidelity error of 6.99 on the testing set, while Circuit B had an error of 6.07.

Steinbrecher et al. [77] introduced Quantum Optical Neural Networks (QONNs), mapping classical neural network concepts to quantum optical systems for tasks like quantum state compression, reinforcement learning, black-box quantum simulation, and one-way quantum repeaters. Using numerical simulations accelerated with Python, Cython, and GPU libraries, QONNs were trained via gradient-free optimization (BOBYQA) and evolutionary strategies, demonstrating strong generalization from limited training data. In a reinforcement learning task, a QRAM-encoded QONN achieved a mean fitness of 136.1 after 1000 generations, which was higher than both a direct-encoded QONN (61.9) and a classical network (37.1). The QONN also successfully compressed molecular hydrogen ground states, with iterative and global optimizations converging to fidelities of 90.0% and 92.2%, respectively. A QONN trained to implement a one-way quantum repeater reached numerical precision after 50 layers.

*c) Quantum chemistry.:* Semi-restricted Quantum Boltzmann Machines (QBMs) have been trained on transverse-field Ising Hamiltonians to approximate correlated-electron ground states. Amin et al. [10] trained Quantum Boltzmann Machines (QBMs) with nondiagonal Hamiltonians using a lower bound on log-likelihood to enable sampling-based gradient estimation. The methodology involved training QBMs with a transverse field Ising Hamiltonian, which introduced quantum effects, and comparing the results to classical Boltzmann machines (BMs) on small systems. The experimental results demonstrated that QBMs consistently learned data distributions better than classical BMs, as measured by lower Kullback-Leibler (KL) divergence values. For a fully visible model with 10 qubits, the classical BM converged to a KL divergence of approximately 0.62, while the QBM achieved a significantly lower value of about 0.42. Table V summarizes all the applications of Quantum GAI models.

## VI. Challenges and Solutions

While the previous sections explored the promise and early applications of quantum generative AI, the path toward its widespread adoption faced significant challenges. These obstacles stemmed from the fundamental limitations of near-term quantum hardware, including issues with scalability, noise, and training stability. The hybrid nature of QGAI also introduced major bottlenecks related to data encoding and the efficient integration of quantum and classical systems. This section examined these core challenges and looked at the future research directions aimed at surmounting them.

### A. Scalability and Hardware

Despite promising early results, quantum generative AI (QGAI) remains fundamentally constrained by the physical limitations of near-term quantum hardware. Current noisy intermediate-scale quantum (NISQ) devices typically provide only 50–100 usable qubits, many of which are prone to decoherence, cross-talk, and gate errors [6], [7]. This restricts the feasible depth and width of parameterized quantum circuits (PQCs), making it difficult to scale quantum generative models for high-dimensional data. Furthermore, existing qubit topologies often impose connectivity constraints that hinder entanglement-efficient circuit designs. As a result, many QGAI frameworks rely on shallow circuits or hybrid quantum-classical workflows to remain viable. To enable large-scale generative modeling, significant advances in qubit fidelity, interconnect density, and error rates are essential. Emerging hardware platforms such as silicon spin qubits, photonic processors, and topological qubits offer promising avenues to transcend current limitations, though they are still in early stages of development [78].

### B. Noise and Error

Quantum noise remains one of the most critical barriers to effective training and deployment of QGAI models. Decoherence, gate infidelity, and measurement noise not only reduce output fidelity but also disrupt gradient estimation during optimization, which is central to variational training techniques like QGANs and QVAEs.

While error-correcting codes are the gold standard in fault-tolerant quantum computing, their overhead remains prohibitive for most current devices. Consequently, variational error mitigation (VEM), zero-noise extrapolation, and probabilistic error cancellation are emerging as more practical approaches for near-term generative tasks [7], [79]. Tailoring these techniques specifically for generative architectures,such as denoising diffusion models or adversarial networks,requires careful trade-offs between circuit expressivity and robustness. Moreover, advances in error-aware training algorithms, noise-resilient ansätze, and active feedback from classical co-processors may play a pivotal role in stabilizing training dynamics under real-world noise conditions.

### C. Training Stability and Barren Plateaus

A major impediment to training quantum generative models is the presence of *barren plateaus*, regions in the optimization landscape where gradients vanish exponentially with circuit depth or system size [13]. This issue severely

Table V: Applications of QGAI models

| Domain | Description |
|---|---|
| Human-machine interaction | QGAI enhances NLP and multimodal interaction, enabling real-time adaptivity and culturally nuanced translations. Example: Quantum Computational Intelligence (QCI) agent integrating ontology, GAI, SBERT similarity, and speech translation for bilingual education [63]. |
| Drug discovery | QGANs and QVAEs encode molecules into quantum states to explore chemical space. Hybrid quantum–classical GANs improve drug-likeness metrics (QED, logP, validity, uniqueness) compared to classical MolGAN [69], [70]. |
| IoT security | QGANs enable anomaly detection with fewer samples than classical GANs [65]. A GAI–blockchain–quantum encryption protocol achieves 99.4% accuracy, 99.1% precision, 98.66% recall, 98.5% F1-score, and 99.2% security in AIoT healthcare [73]. |
| Resource allocation | QGANs and QVAEs with reinforcement learning optimize decentralized resource allocation in quantum networks. Multi-agent RL reduces system costs by at least 80% compared to random allocation [74]. |
| Miscellaneous | Finance: QCBMs simulate stock-market copulas with 98% fidelity; QGANs model cryptocurrency data more efficiently [36], [75].<br>Quantum autoencoders compress quantum states for noise resilience [76].<br>QBMs approximate correlated-electron ground states in quantum chemistry [10]. |

limits the scalability of parameterized quantum circuits (PQCs), especially in high-dimensional generative tasks. Barren plateaus often arise in randomly initialized deep circuits or when cost functions lack locality. Overcoming them requires tailored ansätze with structured entanglement, layer-wise training strategies, or symmetries that preserve gradient signal. Recent research explores locality-aware cost functions, parameter initialization schemes, and adaptive learning-rate schedules to improve convergence [80]. In the context of QGAI, these innovations are particularly crucial since unstable training may degrade sample quality and reduce model fidelity. Exploring quantum-native optimization techniques such as quantum natural gradients or training via quantum feedback loops may further stabilize learning dynamics in generative settings.

### *D. Data Encoding and Readout Bottlenecks*

Bridging the classical–quantum interface is another pressing challenge. Efficiently encoding classical datasets,such as images, molecules, or financial time series,into quantum states is computationally expensive and often circuit-depth intensive. While techniques such as amplitude encoding, basis encoding, and angle encoding are commonly used, each has trade-offs in qubit requirements, expressivity, and hardware overhead [8]. Moreover, measurement readout from quantum circuits is inherently probabilistic and requires repeated sampling to estimate expectation values or reconstruct output distributions, which introduces latency and noise. These bottlenecks become particularly limiting in QGAI pipelines where large volumes of training samples are required. Emerging strategies such as quantum data compression, classical shadows, and low-depth feature maps aim to reduce input/output complexity. Innovations in hybrid encoders and classical pre-processing can also help shift some of the computational burden off quantum hardware [81].

### *E. Resource Requirements and Energy Consumption*

Although quantum algorithms offer potential speedups and energy efficiencies for specific generative tasks, their practical deployment is hampered by the significant resource overhead associated with quantum control systems. Operating qubits at cryogenic temperatures (e.g., below 15 mK) demands elaborate cooling infrastructures, and real-time classical control circuits introduce non-negligible power and synchronization costs. Furthermore, error correction,while essential for fault-tolerant operation,imposes large qubit overheads that dilute energy savings. For instance, implementing a single logical qubit may require hundreds or thousands of physical qubits in surface codes. As quantum workloads scale, the total energy footprint may rival or exceed that of high-performance classical GPU clusters unless more efficient architectures are developed. Designing sustainable, resource-aware quantum-classical systems with optimized thermodynamic and architectural efficiency is thus vital for the widespread adoption of QGAI solutions [81].

### *F. Model Interpretability and Explainability*

As Quantum Generative AI expands from proof-of-concept demos to mission-critical deployments in finance, medicine, and national security, black-box behaviour becomes unacceptable. Classical explainability toolkits,saliency maps, SHAP values, counterfactuals,do not translate directly to the Hilbert space, where information is encoded as complex amplitudes rather than activations. Recent work proposes quantum feature-attribution techniques that combine classical shadows with integrated gradients to reveal which input dimensions dominate a parameterised quantum circuit's output probability mass. Other approaches leverage circuit cutting and local tomography to visualise entanglement flow or to extract low-rank latent representations that correlate with human-interpretable factors [15]. Developing lightweight, hardware-friendly diagnostics that scale beyond a few dozen

qubits is therefore pivotal if QGAI is to earn stakeholder trust in high-stakes settings.

### G. Integration with Classical AI Systems

Most near-term deployments will remain hybrid: a quantum co-processor handles the generative core, while classical hardware pre-processes data and post-processes measurement outcomes. Unfortunately, data marshaling between the host CPU/GPU and the QPU incurs latencies that can outweigh quantum acceleration gains. State-of-the-art orchestration frameworks such as Qiskit Runtime, Amazon Braket Hybrid Jobs, and PennyLane Lightning offer asynchronous job queues, batched parameter updates, and just-in-time compilation, yet scheduling, memory management, and co-optimisation heuristics remain immature [22]. Future research must focus on zero-copy data paths, streaming encoders that amortise I/O cost over many shots, and compiler passes that fuse classical preprocessing with quantum kernels to minimise turnaround time and energy footprint.

### H. Benchmarking and Standardisation

Progress in QGAI is hampered by a patchwork of bespoke datasets, loss metrics, and reporting conventions. Directly porting ImageNet or CIFAR to quantum form is unrealistic, while small synthetic benchmarks mask scalability issues. The community therefore needs consensus suites that span discrete, continuous, and quantum-native data, along with fidelity, diversity, robustness, and energy metrics [39].

Initiatives such as QMLBench[2] for classification are nascent, but broader participation is required to ensure fair cross-model comparison, reproducibility, and transparent reporting of qubit counts, circuit depths, and hardware noise parameters.

### I. Quantum Advantage and Real-World Applications

Although small-scale demonstrations of quantum supremacy have appeared in sampling tasks [6], [82], a *practical* quantum advantage in generative modelling remains unproven. Establishing such an edge requires carefully chosen application niches where (i) the target distribution is classically intractable, (ii) the quantum circuit depth remains within NISQ limits, and (iii) the end-to-end latency, including data transfer, outperforms highly optimised GPU pipelines. Near-term candidates include molecular conformer generation for lead optimisation in drug discovery, Monte-Carlo copula simulation for tail-risk estimation in finance, and rapid lattice sampling for catalyst design in materials science [14]. Benchmarks on trapped-ion and photonic hardware already show order-of-magnitude speedups in small molecular subspaces [83]. Collaborative efforts between domain specialists, quantum hardware vendors, and cloud-service providers are therefore essential to deliver compelling real-world showcases of quantum-enabled generative AI.

Table VI summarizes the key challenges faced by QGAI models along with the corresponding proposed solutions.

[2]https://github.com/XanaduAI/qml-benchmarks

## VII. Future Directions

To further advance the field, we identified some future discussions focused on algorithmic innovation, hybrid orchestration, and trustworthy QGAI as discussed below:

### A. Algorithmic Innovations and New Paradigms

Beyond hardware progress, sustained advances in quantum algorithms will be pivotal for overcoming present-day bottlenecks. Quantum diffusion models extend classical score-based and denoising diffusion frameworks by replacing neural score networks with parameterised quantum circuits that denoise in Hilbert space, promising exponentially richer latent dynamics [11], [12]. Meanwhile, quantum-aware reinforcement learning (QRL) integrates quantum policy networks and quantum environment simulators, allowing agents to exploit entanglement-enhanced exploration and potentially quadratic sample-complexity reductions. Early studies report faster convergence on combinatorial-optimization benchmarks and improved robustness to stochasticity [84]. Other frontiers include quantum meta-learning, quantum natural-gradient descent, and adaptive ansätze that co-evolve with hardware noise profiles. Taken together, these innovations expand the design space of QGAI well beyond direct porting of classical architectures, paving the way for hybrid models that can harness the best of both quantum and classical worlds. Table VI summarizes the key challenges faced by QGAI models along with the corresponding proposed solutions.

### B. Hybrid Orchestration, Compiler Co-Design, and Systems Integration

A critical direction is the co-design of hybrid runtime stacks that minimise host–device latency while maximising quantum utilisation. Promising avenues include batched parameter-shift execution, just-in-time transpilation with noise-aware gate selection, zero-copy pathways between CPU/GPU memory and QPU buffers, and streaming encoders that amortise state preparation over many shots. Compiler passes that fuse classical pre- and post-processing with quantum kernels can reduce I/O overhead and enable end-to-end auto-differentiation across heterogeneous boundaries. At the architectural level, topology-aware ansätze and placement strategies that respect device connectivity constraints can lower circuit depth, while schedule co-optimisation across classical accelerators and QPUs targets wall-clock time, throughput, and energy per generated sample as joint objectives.

### C. Error Mitigation, Verification, and Trustworthy QGAI

Near-term deployment requires techniques that stabilise training and provide statistical guarantees about generated outputs under noise. Variational error-mitigation protocols, zero-noise extrapolation, and probabilistic error cancellation should be tailored to generative losses and sampling workflows. Complementary verification tools, such as shadow-based estimators for distributional properties, robust hypothesis tests for mode collapse, and noise-aware

Table VI: Challenges and Proposed Solutions in Quantum Generative AI (QGAI)

| **Challenge** | **Description** | **Proposed Solutions / Approaches** |
|---|---|---|
| Scalability and hardware limitations | NISQ devices are prone to decoherence, cross-talk, and gate errors. Limits circuit depth, width, and entanglement-efficient designs [7]. | HQC(Hybrid Quantum Classical) systems address NISQ limitations by delegating the computationally difficult parts of a problem to the quantum computer while a classical computer handles the remaining tasks [85]. |
| Noise and error mitigation | Decoherence, gate infidelity, and measurement noise reduce fidelity and disrupt gradient estimation during variational training [7]. | Variational error mitigation, zero-noise extrapolation, probabilistic error cancellation, noise-resilient ansätze, and active feedback [79]. |
| Training stability and barren plateaus | Gradients vanish exponentially with depth or system size, limiting PQC scalability in high-dimensional tasks [13]. | Structured ansätze, layer-wise training, locality-aware cost functions, adaptive learning-rate schedules, and quantum natural gradients [80]. |
| Data encoding and readout bottlenecks | Classical-to-quantum encoding (amplitude, basis, angle) is costly in qubits and depth. Readout requires repeated sampling, adding latency and noise [8]. | Quantum data compression, classical shadows, low-depth feature maps, hybrid encoders, and classical pre-processing [81]. |
| Resource requirements and energy consumption | Cryogenic cooling and control electronics add overhead; error correction requires hundreds of physical qubits per logical qubit. Energy demands may rival GPU clusters. | Sustainable, resource-aware quantum–classical systems; efficient hardware architectures to reduce thermodynamic cost [81]. |
| Model interpretability and explainability | Black-box behaviour in finance, medicine, and security is problematic; Hilbert-space amplitudes resist classical explainability tools. | Quantum feature attribution (classical shadows + gradients), circuit cutting, and local tomography for entanglement flow [15]. |
| Integration with classical AI systems | Hybrid deployment requires data transfer between CPUs/GPUs and QPUs, incurring latency and overhead. | Orchestration frameworks (Qiskit Runtime, Braket Hybrid Jobs, PennyLane Lightning) with asynchronous queues and just-in-time compilation [22]. |
| Benchmarking and standardisation | Lack of standard datasets and metrics; synthetic benchmarks hide scalability issues. | Suites like QMLBench with fidelity, diversity, robustness, and energy metrics for fair comparison. |
| Quantum advantage and applications | Quantum advantage in generative modeling remains unproven; small-scale supremacy demos exist only in sampling tasks [6], [82]. | Domain-focused tasks (drug discovery, finance, materials science), benchmarking on trapped-ion and photonic hardware [14], [83]. |
| Algorithmic innovations and paradigms | Need for quantum-native algorithms beyond classical analogues. | Quantum diffusion models [11], [12], quantum RL with entanglement-enhanced exploration [84], meta-learning, and adaptive ansätze. |

calibration loops, can quantify fidelity and diversity without prohibitive measurement budgets. Interpretability remains central: feature-attribution in Hilbert space, circuit-cutting visualisations of entanglement flow, and sensitivity analyses for data embeddings can help diagnose failure modes and build practitioner trust in safety-critical domains.

### *D. Data Encoding, Compression, and Quantum-Native Corpora*

The classical–quantum interface will benefit from learned and problem-specific embeddings that reduce state-preparation cost while retaining sufficient statistics for generation. Future work should explore variational data loaders, quantum-assisted tokenisation for sequences, and compression schemes that couple classical autoencoders with shallow quantum layers to trade qubits for classical latent structure. On the readout side, grouped measurements, shot-frugal estimators, and adaptive stopping rules can lower sampling requirements during training and evaluation. Community benchmarks comprising quantum-native datasets, along with standardised protocols for encoding images, molecules, and time-series, will enable fair comparisons across architectures and hardware backends.

### *E. Benchmarking, Reproducibility, and Evidence for Advantage*

To move beyond proof-of-concept demonstrations, the field needs principled benchmarks that report fidelity, diversity, robustness to distribution shift, energy per sample, and end-to-end wall-clock time on specified hardware. Reproducible evaluation suites with fixed encoders, circuit-depth budgets, and noise profiles can disentangle algorithmic progress from hardware drift. Application-driven targets, molecular conformer generation, tail-risk copula sampling, and materials lattice sampling, are well-suited for establishing compelling case studies of practical advantage when combined with transparency about qubit counts, connectivity, calibration data, and compiler settings. Public leaderboards and standard result cards should accompany releases to encourage rigorous, apples-to-apples assessment across QGAI systems.

## VIII. Conclusions

Quantum Generative AI (QGAI) combines the expressive power of generative models with the computational advantages of quantum mechanics, offering a promising pathway beyond the limitations of classical AI. By harnessing principles like superposition and entanglement, QGAI architectures can represent and generate complex data distributions more efficiently than their classical counterparts. This paper has highlighted the core frameworks such as QCBMs, QVAEs, and Quantum Diffusion Models and their applications in domains like drug discovery, IoT security, and human–machine interaction. While current progress is constrained by hardware noise, limited qubit counts, and algorithmic challenges like barren plateaus, the field is advancing rapidly. With continued improvements in quantum hardware and hybrid learning frameworks, QGAI is well-positioned to contribute significantly to future intelligent systems. Its potential to enable scalable, efficient, and privacy-preserving generative modeling makes it a compelling direction for ongoing research and development, which are highlighted in the paper.

## Acknowledgements

This work is supported by CHANAKYA Fellowship Program of TIH Foundation for IoT & IoE (TIH-IoT) received by Dr. Vinay Chamola under Project Grant File CFP/2022/027 and Melbourne University Visiting stipend.

## References

[1] Z. Ding, Y. Xu, W. Xu, G. Parmar, Y. Yang, M. Welling, and Z. Tu, "Guided variational autoencoder for disentanglement learning," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 7920–7929.

[2] P. Dhariwal and A. Nichol, "Diffusion models beat gans on image synthesis," in *Advances in Neural Information Processing Systems*, vol. 34, 2021, pp. 8780–8794.

[3] T. Brown, B. Mann, N. Ryder, M. Subbiah, Kaplan *et al.*, "Language models are few-shot learners," in *Advances in Neural Information Processing Systems*, 2020.

[4] R. Bommasani *et al.*, "On the opportunities and risks of foundation models," *arXiv:2108.07258*, 2021.

[5] D. Patterson, J. Gonzalez, Q. Le, C. Liang, L. Munguia, D. Rothchild, D. So, M. Texier, and J. Dean, "Carbon emissions and large neural network training," *arXiv:2104.10350*, 2021.

[6] F. Arute, K. Arya, R. Babbush, D. Bacon, J. C. Bardin, R. Barends, R. Biswas *et al.*, "Quantum supremacy using a programmable superconducting processor," *Nature*, vol. 574, pp. 505–510, 2019.

[7] J. Preskill, "Quantum computing in the nisq era and beyond," *Quantum*, vol. 2, p. 79, 2018.

[8] M. Benedetti, E. Lloyd, S. Sack, and M. Fiorentini, "Parameterized quantum circuits as machine learning models," *Quantum Science and Technology*, vol. 4, no. 4, p. 043001, 2019.

[9] S. Lloyd and C. Weedbrook, "Quantum generative adversarial learning," *Physical Review Letters*, vol. 121, no. 4, p. 040502, 2018.

[10] M. H. Amin, E. Andriyash, J. Rolfe, B. Kulchytskyy, and R. Melko, "Quantum boltzmann machine," *Physical Review X*, vol. 8, no. 2, p. 021050, 2018.

[11] B. Zhang, P. Xu, X. Chen, and Q. Zhuang, "Generative quantum machine learning via denoising diffusion probabilistic models," *Physical Review Letters*, vol. 132, p. 100602, 2024.

[12] C. Chen, Q. Zhao, M. Zhou, Z. He, Z. Sun, and H. Situ, "Quantum generative diffusion model: a fully quantum-mechanical model for generating quantum state ensemble," *arXiv:2401.07039*, 2024.

[13] M. Cerezo, A. Sone, T. Volkoff, L. Cincio, and P. J. Coles, "Trainability barriers and opportunities in quantum generative modeling," *npj Quantum Information*, vol. 10, no. 1, p. 16, 2024.

[14] M. Benedetti, D. Garcia-Pintos, O. Perdomo, Y. Nam, and A. Perdomo-Ortiz, "A framework for demonstrating practical quantum advantage," *Nature Communications*, vol. 15, no. 1, p. 1552, 2024.

[15] X. Gao, E. R. Anschuetz, S.-T. Wang, J. I. Cirac, and M. D. Lukin, "Enhancing generative models via quantum correlations," *Physical Review X*, vol. 12, no. 2, p. 021037, 2022.

[16] J. Preskill, "Are quantum computers really energy efficient?" *Nature Reviews Physics*, vol. 4, pp. 362–363, 2022.

[17] Y. Zhao, H. Wang, K. Xu, Y. Wang, J. Zhu, and F. Wang, "Adaptive algorithm for quantum amplitude estimation," *Entropy*, vol. 24, no. 8, p. 1015, 2022.

[18] A. Bacho, H. Boche, and G. Kutyniok, "Reliable ai: Does the next generation require quantum computing?" *arXiv:2307.01301*, 2023.

[19] G. Acampora, A. Ambainis, N. Ares, L. Banchi, P. Bhardwaj *et al.*, "Quantum computing and artificial intelligence: status and perspectives," arXiv:2505.23860, 2025.

[20] I. Fernández Pérez, F. de la Prieta, S. Rodríguez-González, J. M. Corchado, and J. Prieto, "Quantum ai: Achievements and challenges in the interplay of quantum computing and artificial intelligence," in *ISAmI 2022*, ser. Lecture Notes in Networks and Systems. Cham: Springer Nature Switzerland, 2023, vol. 603, pp. 155–166.

[21] A. Rayhan and S. Rayhan, "Quantum computing and ai: A quantum leap in intelligence," *AI Odyssey: Unraveling the Past, Mastering the Present, and Charting the Future of Artificial Intelligence*, vol. 424, 2023.

[22] M. Jindal, "Enhancing generative ai models through quantum computing for improved text and image generation," *SSRN*, no. 5198542, 2025.

[23] I. J. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, and Y. Bengio, "Generative adversarial nets," in *Advances in Neural Information Processing Systems (NeurIPS 27)*, 2014, pp. 2672–2680.

[24] D. P. Kingma and M. Welling, "Auto-encoding variational bayes," arXiv:1312.6114, 2013.

[25] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems*, 2017.

[26] L. Ouyang, J. Wu, X. Jiang, D. Almeida, C. Wainwright, P. Mishkin, C. Zhang, S. Agarwal, K. Slama, A. Ray *et al.*, "Training language models to follow instructions with human feedback," *Advances in neural information processing systems*, vol. 35, pp. 27 730–27 744, 2022.

[27] J. Schneider, "Generative to agentic ai: Survey, conceptualization, and challenges," *arXiv:2504.18875*, 2025.

[28] P. W. Shor, "Algorithms for quantum computation: discrete logarithms and factoring," in *Proceedings 35th annual symposium on foundations of computer science*. IEEE, pp. 124–134, 1994.

[29] L. K. Grover, "A fast quantum mechanical algorithm for database search," in *Proceedings of the twenty-eighth annual ACM symposium on Theory of computing*, pp. 212–219, 1996.

[30] M. AbuGhanem, "Superconducting quantum computers: who is leading the future?" *EPJ Quantum Technology*, vol. 12, no. 1, p. 102, 2025.

[31] C. D. Bruzewicz, J. Chiaverini, R. McConnell, and J. M. Sage, "Trapped-ion quantum computing: Progress and challenges," *Applied physics reviews*, vol. 6, no. 2, 2019.

[32] M. AbuGhanem, "Photonic quantum computers," *arXiv:2409.08229*, 2024.

[33] J. Wurtz, A. Bylinskii, B. Braverman, J. Amato-Grill, S. H. Cantu, F. Huber, A. Lukin, F. Liu, P. Weinberg, J. Long *et al.*, "Aquila: Quera's 256-qubit neutral-atom quantum computer," *arXiv:2306.11727*, 2023.

[34] R. P. Feynman, "Simulating physics with computers," *International Journal of Theoretical Physics*, vol. 21, no. 6, pp. 467–488, 1982.

[35] J.-G. Liu and L. Wang, "Differentiable learning of quantum circuit born machines," *Physical Review A*, vol. 98, no. 6, p. 062324, 2018.

[36] D. Zhu, N. M. Linke, M. Benedetti, K. A. Landsman *et al.*, "Training of quantum circuits on a hybrid quantum computer," *Science Advances*, vol. 5, no. 9, 2019.

[37] M. S. Daoud, M. Shehab, H. M. Al-Mimi, L. Abualigah, R. Abu Zitar, and M. K. Y. Shambour, "Gradient-based optimizer (gbo): a review, theory, variants, and applications," *Archives of Computational Methods in Engineering*, vol. 30, no. 4, pp. 2431–2449, 2023.

[38] X. Li and X. Yao, "Cooperatively coevolving particle swarms for large scale optimization," *IEEE Transactions on Evolutionary Computation*, vol. 16, no. 2, pp. 210–224, 2011.

[39] C. A. Riofrio, O. Mitevski, C. Jones, F. Krellner, A. Vuckovic, J. Doetsch, J. Klepsch, T. Ehmer, and A. Luckow, "A characterization of quantum generative models," *ACM Transactions on Quantum Computing*, vol. 5, no. 2, pp. 1–34, 2024.

[40] L.-H. Gong, L.-Z. Xiang, S.-H. Liu, and N.-R. Zhou, "Born machine model based on matrix product state quantum circuit," *Physica A: Statistical Mechanics and its Applications*, vol. 593, p. 126907, 2022.

[41] W. Huggins, P. Patil, B. Mitchell, K. B. Whaley, and E. M. Stoudenmire, "Towards quantum machine learning with tensor networks," *Quantum Science and Technology*, vol. 4, no. 2, p. 024001, 2019.

[42] H.-L. Huang, Y. Du, M. Gong, Y. Zhao, Y. Wu, C. Wang, S. Li, J. Li, F. Liang, J. Lin, Y. Xu, Z. Guo, L. Guo, J. Zhao, F. Zhong, Y. Zhang, C.-Y. Wang, X.-H. Peng, H. Zhang, H. Xu, L. He, X. Chen, Q. Luo, C. Chen, Y. Wang, Y. Ding, and J.-W. Pan, "Experimental quantum generative adversarial networks for image generation," *Physical Review Applied*, vol. 16, no. 2, p. 024051, 2021.

[43] S. Chakrabarti, H. Yiming, T. Li, S. Feizi, and X. Wu, "Quantum wasserstein generative adversarial networks," *Advances in Neural Information Processing Systems*, vol. 32, 2019.

[44] B. T. Kiani, G. De Palma, M. Marvian, Z.-W. Liu, and S. Lloyd, "Learning quantum data with the quantum earth mover's distance," *Quantum Science and Technology*, vol. 7, no. 4, p. 045002, 2022.

[45] M. Schuld, V. Bergholm, C. Gogolin, J. Izaac, and N. Killoran, "Evaluating analytic gradients on quantum hardware," *Physical Review A*, vol. 99, no. 3, p. 032308, 2019.

[46] M. Kieferová and N. Wiebe, "Tomography and generative data modeling via quantum boltzmann training," *Physical Review A*, vol. 96, no. 6, p. 062327, 2017.

[47] D. Crawford, A. Levit, N. Ghadermarzy, J. S. Oberoi, and P. Ronagh, "Reinforcement learning using quantum boltzmann machines," *arXiv:1612.05695*, 2016.

[48] C. Zoufal, A. Lucchi, and S. Woerner, "Variational quantum boltzmann machines," *Quantum Machine Intelligence*, vol. 3, no. 1, p. 7, 2021.

[49] M. Minervini, D. Patel, and M. M. Wilde, "Evolved quantum boltzmann machines," *arXiv:2501.03367*, 2025.

[50] L. Coopmans and M. Benedetti, "On the sample complexity of quantum boltzmann machine learning," *Communications Physics*, vol. 7, no. 1, p. 274, 2024.

[51] A. Khoshaman, W. Vinci, B. Denis, E. Andriyash, H. Sadeghi, and M. H. Amin, "Quantum variational autoencoder," *Quantum Science and Technology*, vol. 4, no. 1, p. 014001, 2018.

[52] C. Bravo-Prieto, "Quantum autoencoders with enhanced data encoding," *Machine Learning: Science and Technology*, vol. 2, no. 3, p. 035028, 2021.

[53] M. Srikumar, C. D. Hill, and L. C. L. Hollenberg, "Clustering and enhanced classification using a hybrid quantum autoencoder," *Quantum Science and Technology*, vol. 7, no. 1, p. 015020, 2021.

[54] C. Cao and X. Wang, "Noise-assisted quantum autoencoder," *Physical Review Applied*, vol. 15, no. 5, p. 054012, 2021.

[55] G. Wang, J. Warrell, P. S. Emani, and M. Gerstein, "$\zeta$-QVAE: A Quantum Variational Autoencoder utilizing Regularized Mixed-state Latent Representations," *arXiv:2402.17749*, 2024.

[56] A. Cacioppo, L. Colantonio, S. Bordoni, and S. Giagu, "Quantum diffusion models," arXiv:2311.15444, 2023.

[57] M. Kölle, G. Stenzel, J. Stein, S. Zielinski, and C. Linnhoff-Popien, "Quantum denoising diffusion models," arXiv:2401.07049, 2024.

[58] V. Dakulagi, K. H. Yeap, H. Nisar, R. Dakulagi, G. N. Basavaraj, and M. Villagómez Galindo, "An overview of techniques and best practices to create intuitive and user-friendly human-machine interfaces," in *Artificial Intelligence and Multimodal Signal Processing in Human-Machine Interaction*. Springer, 2025, pp. 63–77.

[59] Y. Kang, Z. Cai, C.-W. Tan, Q. Huang, and H. Liu, "Natural language processing (nlp) in management research: A literature review," *Journal of Management Analytics*, vol. 7, no. 2, pp. 139–172, 2020.

[60] A. Joshi, R. Dabre, D. Kanojia, Z. Li, H. Zhan, G. Haffari, and D. Dippold, "Natural language processing for dialects of a language: A survey," *ACM Computing Surveys*, vol. 57, no. 6, pp. 1–37, 2025.

[61] I. Grobelna, D. Mailland, and M. Horwat, "Design of automotive hmi: New challenges in enhancing user experience, safety, and security," *Applied Sciences*, vol. 15, no. 10, p. 5572, 2025.

[62] Z. Albassami, A. Algarni, A. Qahmash, and Z. Ahmad, "A comprehensive review of ai-driven q&a systems with taxonomy, prospects, and challenges," *Knowledge and Information Systems*, pp. 1–24, 2025.

[63] C.-S. Lee, M.-H. Wang, J.-K. Chiang, N. Kubota, E. Sato-Shimokawara, Y. Nojima, G. Acampora *et al.*, "Quantum computational intelligence with generative ai image for human-machine interaction," in *Proceedings of the 2024 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE)*. IEEE, pp. 1–8, 2024.

[64] A. V. Sadybekov and V. Katritch, "Computational approaches streamlining drug discovery," *Nature*, vol. 616, no. 7958, pp. 673–685, 2023.

[65] N. Berdigaliyev and M. Aljofan, "An overview of drug discovery and development," *Future medicinal chemistry*, vol. 12, no. 10, pp. 939–947, 2020.

[66] Y. A. Ivanenkov, D. Polykovskiy, D. Bezrukov, B. Zagribelnyy, V. Aladinskiy, P. Kamya, A. Aliper, F. Ren, and A. Zhavoronkov, "Chemistry42: an ai-driven platform for molecular design and optimization," *Journal of chemical information and modeling*, vol. 63, no. 3, pp. 695–701, 2023.

[67] S. A. Stein, B. Baheri, D. Chen, Y. Mao, Q. Guan, A. Li, B. Fang, and S. Xu, "Qugan: A quantum state fidelity based generative adversarial network," in *Proceedings of the 2021 IEEE International Conference on Quantum Computing and Engineering (QCE)*. IEEE, pp. 71–81, 2021.

[68] G. R. Bickerton, G. V. Paolini, J. Besnard, S. Muresan, and A. L. Hopkins, "Quantifying the chemical beauty of drugs," *Nature Chemistry*, vol. 4, pp. 90–98, 2012.

[69] J. Li and S. Ghosh, "Scalable variational quantum circuits for autoencoder-based drug discovery," in *2022 design, automation & test in europe conference & exhibition (DATE)*. IEEE, pp. 340–345, 2022.

[70] P. Jain and S. Ganguly, "Hybrid quantum generative adversarial networks for molecular simulation and drug discovery," *arXiv:2212.07826*, 2022.

[71] J. Xiong, L. Shen, Y. Liu, and X. Fang, "Enhancing iot security in smart grids with quantum-resistant hybrid encryption," *Scientific Reports*, vol. 14, p. 84427, 2024.

[72] E. Bermot, C. Zoufal, M. Grossi, J. Schuhmacher, F. Tacchino, S. Vallecorsa, and I. Tavernelli, "Quantum generative adversarial networks for anomaly detection in high energy physics," in *Proceedings of the 2023 IEEE International Conference on Quantum Computing and Engineering (QCE)*. IEEE, 2023, pp. 331–341.

[73] S. Prajapat, P. Kumar, A. K. Das, and G. Muhammad, "Generative ai-enabled quantum encryption algorithm for securing iot-based healthcare application using blockchain," *IEEE Internet of Things Journal*, vol. 12, no. 13, pp. 24 541–24 551, 2025.

[74] M. Xu, D. Niyato, J. Kang, Z. Xiong, Y. Cao, Y. Gao, C. Ren, and H. Yu, "Generative ai-enabled quantum computing networks and intelligent resource allocation," *arXiv:2401.07120*, 2024.

[75] S. Ganguly, "Implementing quantum generative adversarial network (qgan) and qcbm in finance," *arXiv:2308.08448*, 2023.

[76] J. Romero, J. P. Olson, and A. Aspuru-Guzik, "Quantum autoencoders for efficient compression of quantum data," *Quantum Science and Technology*, vol. 2, no. 4, p. 045001, 2017.

[77] G. R. Steinbrecher, J. P. Olson, D. Englund, and J. Carolan, "Quantum optical neural networks," *npj Quantum Information*, vol. 5, p. 60, 2019.

[78] B. Lekitsch, S. Weidt, A. G. Fowler, K. Mølmer, S. J. Devitt, C. Wunderlich, and W. K. Hensinger, "Blueprint for a microwave trapped ion quantum computer," *Science Advances*, vol. 3, no. 2, p. e1601540, 2017.

[79] K. Temme, S. Bravyi, and J. M. Gambetta, "Error mitigation for short-depth quantum circuits," *Physical Review Letters*, vol. 119, no. 18, p. 180509, 2017.

[80] J. R. McClean, S. Boixo, V. N. Smelyanskiy, R. Babbush, and H. Neven, "Barren plateaus in quantum neural network training landscapes," *Nature Communications*, vol. 9, p. 4812, 2018.

[81] H.-Y. Huang, R. Kueng, and J. Preskill, "Efficient estimation of quantum observables with classical shadows," *Nature Physics*, vol. 16, pp. 1050–1057, 2020.

[82] H.-S. Zhong, H. Wang, Y.-H. Deng, and et al., "Quantum computational advantage using photons," *Science*, vol. 370, no. 6523, pp. 1460–1463, 2020.

[83] L. S. Madsen, F. Laudenbach, M. F. Askarani, F. Rortais, T. Vincent *et al.*, "Quantum computational advantage with a programmable photonic processor," *Nature*, vol. 606, pp. 75–81, 2022.

[84] Y. Hu, F. Tang, J. Chen, and W. Wang, "Quantum-enhanced reinforcement learning for control: A preliminary study," *Control Theory and Technology*, vol. 19, pp. 455–464, 2021.

[85] G. De Luca, "A survey of NISQ era hybrid quantum-classical machine learning research," *Journal of Artificial Intelligence and Technology*, vol. 2, no. 1, pp. 9–15, 2022.